Qi Zhang

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Date of Birth: October 30, 1995

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Research Profile

Computational mechanics researcher working on nonlinear finite element (FE) dynamics of flexible space systems. My doctoral work turned high-fidelity FE models into the direct basis for state estimation and controller synthesis, without first reducing them to low-order surrogates. I am now extending this work toward quantum algorithms for nonlinear FE dynamics through Koopman lifting. Author of twelve journal papers (ten published, two under review), including seven as first author and two as co-first author, and four conference presentations.

Education

Ph.D., Space Science and Engineering 2021 – 2026

York University, Department of Earth and Space Science and Engineering *Toronto, Canada*

- Thesis: *From computational mechanics to computational control: automated control synthesis for elastic tethered space systems*
- Supervisor: Prof. Zheng H. Zhu

Master, Mechanics 2018 – 2021

Shanghai Jiao Tong University, School of Naval Architecture, Ocean and Civil Engineering *Shanghai, China*

- Thesis: *Dynamic Modeling and Active Control for Space Robots Considering Joint Flexibility and Friction*
- Supervisor: Prof. Guoping Cai

Bachelor, Engineering Mechanics 2014 – 2018

Dalian University of Technology, Faculty of Vehicle Engineering and Mechanics *Dalian, China*

Research Experience

Postdoctoral Fellow Jul. 2026 – Present

York University, Department of Mechanical Engineering *Toronto, Canada*

- Postdoctoral research in space engineering under Prof. Zheng H. Zhu.
- Developing physics-aware quantum algorithms for nonlinear FE dynamics through Koopman lifting.

Doctoral Student 2021 – 2026

York University, Department of Earth and Space Science and Engineering *Toronto, Canada*

- Developed a unified nodal-position FE model of spacecraft and tethers that captures orbital–structural coupling.
- Built an FE–Kalman estimator that reconstructs full tether states from sparse position, velocity and fibre-optic strain measurements.
- Designed system-level and element-level computational controllers synthesized directly on full-order FE models.

Master Student 2018 – 2021
Shanghai Jiao Tong University *Shanghai, China*

- Modelled and controlled flexible-link, flexible-joint space robots with joint friction, from formulation through numerical validation to journal publication [J11].

Undergraduate Student 2014 – 2018
Dalian University of Technology *Dalian, China*

- Projects in analytical mechanics, contact dynamics, FE analysis and vibration isolation; co-authored a study of point-supported thin plates [J12].

Key Research Contributions

1. Computational control synthesized directly on high-fidelity FE dynamics

Developed a framework in which full-order FE dynamics serve as the basis for state estimation and controller synthesis, not only for post-design simulation. This avoids reduced-order surrogates that suppress distributed deformation, high-frequency dynamics and internal-force interactions.

- *System-level FE optimal control:* formulated optimal control on the assembled FE equations, converted the Hamiltonian optimality conditions into a solvable two-point boundary-value problem, and added moving-horizon updates with FE–Kalman feedback.
- *Element-level co-located control:* synthesized local control laws at the element level and assembled them into a global feedback law through FE connectivity, with stability established by Lyapunov analysis.
- *Impact:* a complete modelling–estimation–control chain within one computational mechanics framework, opening a route to automated control synthesis for high-dimensional flexible systems.
- *Representative outputs:* [J6], [J4], [C3], [C4].

2. Unified FE modelling of coupled orbital and structural dynamics

Used the Nodal Position Finite Element Method (NPFEM) to represent large rigid-body motion, orbital motion and structural deformation in one mechanically consistent state description.

- *NPFEM for rotating elastic solids:* captured large-motion elastodynamics, centrifugal stiffening and nonlinear vibration within a single FE representation.
- *Full-elastic tethered spacecraft:* modelled spacecraft and tethers as compatible elastic elements, retaining tether tension, connection loads and orbital–structural coupling; extended to the deployment of spinning hub–spoke formations with an Arbitrary Lagrange–Euler NPFEM.
- *Impact:* the FE model becomes the system dynamics model itself, resolving distributed quantities that simplified rigid–flexible models cannot.
- *Representative outputs:* [J7], [J3], [C2].

3. Distributed FE state estimation from sparse measurements

Developed an FE–Kalman framework that reconstructs the distributed tether state from a limited set of realistic measurements.

- *Method:* fuses endpoint position and velocity, distributed fibre-optic strain sensing, inverse FE reconstruction and model-based virtual sensing.
- *Impact:* links physically realizable sensing to the same distributed state used for modelling and control, making the FE representation usable in closed loop.
- *Representative outputs:* [J10], [C3].

Journal Publications

Under Review

[J1] Li, J.[†], **Zhang, Q.**[†], Wong, D. C. Y.[†], Yao, F.[†], Liu, Z.[†], & Zhu, Z. H. (2026). Artificial intelligence in and for space exploration: A landscape review of methods, applications, and challenges. *Progress in Aerospace Sciences*. (Under Review) (Invitation only) **[Q1]**

[J2] **Zhang, Q.**, & Zhu, Z. H. (2026). Tethered space system for near-surface lunar observation. *Journal of Guidance, Control, and Dynamics*. (Under Review)

Published

[J3] **Zhang, Q.**, & Zhu, Z. H. (2026). Orbital-structural coupling effects in tethered space systems using unified elastic dynamic modeling. *Aerospace Science and Technology*, 113105. **[Q1]**

[J4] **Zhang, Q.**, & Zhu, Z. H. (2026). Co-located finite element modeling and computational control for tethered space systems. *Chinese Journal of Aeronautics*, 104336. **[Q1]**

[J5] Yao, F.[†], **Zhang, Q.**[†], & Zhu, Z. H. (2026). Near-surface lunar observation by tethered orbital platform for high-resolution long-duration exploration. *Acta Astronautica*, 246, 778–783. **[Q1]**

[J6] **Zhang, Q.**, & Zhu, Z. H. (2025). Computational control framework for tethered space systems using finite element method. *Acta Astronautica*, 228, 595–606. **[Q1]**

[J7] **Zhang, Q.**, & Zhu, Z. H. (2025). Elastodynamic analysis of rotating solids by novel nodal position finite element method. *European Journal of Mechanics – A/Solids*, 109, 105478. **[Q1]**

[J8] Zhu, W., Liu, Z., **Zhang, Q.**, Pang, Z., & Zhu, Z. H. (2025). Development of pseudo-celestial navigation and tracking system for planar air-bearing satellite simulators. *Aerospace Science and Technology*, 111462. **[Q1]**

[J9] Liu, Z., **Zhang, Q.**, & Zhu, G. Z. (2025). A comprehensive review on computational control. *Transactions of the Canadian Society for Mechanical Engineering*, 50, 1–16.

[J10] **Zhang, Q.**, Asri El Ghali, A., & Zhu, Z. H. (2023). Space tether system state estimation using distributed fibre optical strain measurement, inverse finite element method and Kalman filter. *Acta Astronautica*, 211, 435–446. **[Q1]**

[J11] **Zhang, Q.**, Liu, X., & Cai, G. (2021). Dynamics and control of a flexible-link flexible-joint space robot with joint friction. *International Journal of Aeronautical and Space Sciences*, 22, 415–432.

[J12] Li, R., Wang, B., Lv, Y., **Zhang, Q.**, Wang, H., Jin, F., et al. (2017). New analytic solutions for static problems of rectangular thin plates point-supported at three corners. *Meccanica*, 52, 1593–1600.

[†] Co-first authors. **[Q1]** Journal in the top quartile of its Journal Citation Reports (JCR) subject category.

Conference Presentations

All peer-reviewed oral presentations.

[C1] **Zhang, Q.**, Yao, F., & Zhu, Z. H. (2026). Skimming the Moon: A tethered spacecraft for near-surface lunar observation. *77th International Astronautical Congress (IAC 2026)*, Antalya, Türkiye.

[C2] **Zhang, Q.**, Yao, F., & Zhu, Z. H. (2026). Deployment dynamics of a spinning hub-spoke tethered spacecraft formation using Arbitrary Lagrange–Euler nodal position finite element method. *Fifth International Nonlinear Dynamics Conference (NODYCON 2026)*, Rome, Italy.

[C3] **Zhang, Q.**, & Zhu, Z. H. (2024). Computational control for flexible cable systems with a finite element state estimator. *7th International Conference on Tethers in Space*, Toronto, Canada.

[C4] **Zhang, Q.**, & Zhu, Z. H. (2024). Computational control framework for flexible cable based on finite element methods. *Canadian Society for Mechanical Engineering (CSME) Congress*, Toronto, Canada.

Honours and Awards

2025	Chinese Government Award for Outstanding Self-Financed Students Abroad *York University*	National
2024	Gold Medal (Team Leader), China International College Students' Innovation Competition *York University and Shanghai Jiao Tong University*	International
2021	York Graduate Scholarship *York University*	Institutional
2021	Lassonde Graduate Scholarship *York University*	Institutional
2021	Outstanding Graduate *Shanghai Jiao Tong University*	Provincial
2019	Chinese National Scholarship *Shanghai Jiao Tong University*	National
2018	Outstanding Graduate *Dalian University of Technology*	Institutional

Professional Service and Leadership

- **Managing Editor**, *Acta Astronautica* 2025 – Present
- **Co-Guest Editor**, *Acta Astronautica* 2025 – 2026
- **Reviewer**, *Acta Astronautica*; *Aerospace Science and Technology*; *Nonlinear Dynamics*
- **Conference Secretary**, 7th International Conference on Tethers in Space, Toronto, Canada 2024
- **Team Leader**, student research team, China International College Students' Innovation Competition (Gold Medal). Coordinated the research direction, integrated contributions across team members, and led the project through to final presentation. 2024